

Stock Markets Analytics Zoomcamp

by Pythoninvest.com in collaboration with DataTalks.Club

Module 1

Introduction And Data Sources

Instructor: Ivan Brigida

Picture

- Business Intelligence Analyst at Google
- Educational background in Computer Science, Finance, and Data Analytics
- Founder of [PythonInvest.com](https://pythoninvest.com)
- Retail investor

Agenda

Outstanding Pre-Launch Questions

Questionnaire update

- Country breakdown

- Applying the knowledge
- Course objectives vs. expected outcomes

Introduction: Making Data-Driven Decisions

- The philosophy of data-informed decision-making
- Overview of personal investment opportunities
- Understanding risk versus reward
- Real-world data examples (S&P 500, interest rates, ETFs)

Hands-On: Colab Setup & Initial Data Collection

- Step-by-step guide to setting up Colab for data analysis
- Downloading initial financial datasets via Finance APIs
- Data examples: OHLCV (YFinance), macro data (FRED), news (Massive.com), alternatives (YouTube?)

Vote for questions or submit yours

Join at

<https://qna.dtcdev.click/r/sma>

Docs and Q&A from
Previous years [https://
datatalks.club/docs/courses/
stock-markets-analytics-
zoomcamp/](https://datatalks.club/docs/courses/stock-markets-analytics-zoomcamp/)

LINK: <https://qna.dtcdev.click/r/sma>

(2026) Selected Unanswered Questions from Slido (#SMA)

Q1: can an experienced data scientist move their careers into quant trading with this course? If not, what additional knowledge would be required?

- ☐ Potentially yes, especially if you have a PhD in Math, Physics, etc. Or won a Kaggle competition from [Jane Street](#), [XTX](#) (2026 - [AIMO contest](#)), etc.
- ☐ You can try joining a hedge fund as a data engineer or software engineer and work directly with quants
- ☐ This is outside the scope of this course

Q2: Is it advisable to automate a trading system where you basically on trade on analysis by telegram signal channels?

- ☐ [Front running alert] I have no experience of using Telegram channels signals (from closed/paid groups)
- ☐ Telegram can be used as an easy front-end to the system (no web/mobile programming required)

Q3: For production systems in this area, what's the best programming language? (I understand we'll be using Python for the course)

☐ I haven't seen many production systems in this area, so I can't comment

Q4: Which stack will we use to make the AI agents?

☐ We won't cover AI agents in this course, but you can absolutely integrate it and get a few bonus points

☐ Ivan did a workshop on building an AI agent in June

Q5: Are there anything we should do in the meantime as a plus, until the beginning of the course?

☐ Watch previous year lectures and try running the code

Q6: What are the two free data sources you use?

☐ YFinance and FRED

☐ Will be covered today

Add more questions here: <https://qna.dtcdev.click/r/sma>

<https://app.sli.do/event/59tMPiTDjYJDWqL4UaPXog/live/questions>

Questionnaire

23 August: Updated analysis for
±900 submissions (1.5x from
earlier results)

**[23 August] Updated Countries Breakdown
(n=821 from 60 countries with >1 participants)**

Concentration: top 10 countries (INDIA, USA, CANADA, UK, GERMANY, NIGERIA, KENYA, SPAIN, INDONESIA, PAKISTAN) are 54% of all respondents.

Questionnaire: 'Programming with AI Experience'

(you can speed up every step of your work: data collection, analysis, ML, coding, brainstorming ideas?)

22% of respondents answered 'No AI tools used yet'

Questionnaire: 'DataTalks.Club experience' (n=328)

(previous courses will help you a lot with #SMA Zoomcamp)

40% of respondents took at least 1 course with DataTalks.Club

[23 August] Applying the Knowledge / Investing Idea (n=305)

(good, medium, low coverage in the course)

Top Project Ideas (Part 1)

1. **Algorithmic and automated trading systems or bots (49)** Build trading bots, automated buy/sell systems, rule-based strategies, trading signals, execution systems, or cloud-based automated trading tools. Examples: stock, forex, crypto, options, and local-market trading bots.
2. **Machine learning, statistical modelling, and market forecasting (34)** Predict stock prices, returns, market trends, volatility, market regimes, company performance, or economic movements using ML and time-series models. Common methods mentioned include XGBoost, Random Forest, LSTM, GARCH, regression, clustering, and reinforcement learning.
3. **Stock screeners, market-analysis dashboards, and monitoring tools (32)** Create dashboards or screening platforms that help users analyze stocks, track portfolios, monitor markets, and identify investment opportunities. Potential features include technical indicators, market alerts, visualizations, watchlists, and automated reports.
4. **Portfolio management, optimization, and rebalancing (22)** Develop tools for portfolio construction, diversification, risk-adjusted performance analysis, asset allocation, rebalancing, and retirement or personal wealth management. Examples: Modern Portfolio Theory, factor-based portfolios, ETF portfolio health checks, and dynamic asset allocation.
5. **Backtesting, simulation, and strategy evaluation (18)** Test trading and investment strategies using historical data and evaluate returns, Sharpe ratio, drawdown, volatility, transaction costs, slippage, and benchmark performance. Several responses specifically mentioned walk-forward validation and avoiding look-ahead or survivorship bias.
6. **Fundamental analysis, equity valuation, and company-performance analysis (18)** Analyze companies using financial statements, valuation models, financial ratios, profitability, growth, leverage, liquidity, and business performance. Ideas include DCF, CAPM, Altman Z-Score, equity research, IPO comparison, and automated financial-document analysis.

Top Project Ideas (Part 2)

1. **Financial data pipelines, APIs, and engineering infrastructure (15)** Build systems that collect, clean, transform, and distribute financial data from APIs, websites, Excel, databases, or real-time market feeds. Examples: Python/SQL ETL pipelines, Excel data bridges, low-latency market-data systems, Redis-based pipelines, and Dockerized workflows.
2. **Risk analytics and financial risk modelling (12)** Develop tools for portfolio risk, volatility, credit risk, financial distress, fraud detection, drawdown analysis, liquidity risk, and stress testing. Examples include credit-card fraud detection, agricultural commodity risk, credit-risk prediction, and risk-aware portfolio construction.
3. **News, sentiment, macroeconomic, and alternative-data analysis (12)** Combine market data with news, social sentiment, economic indicators, interest rates, inflation, supply-chain information, weather events, or geopolitical events. Potential applications include event-driven trading, sentiment-based signals, and assessing the effect of wars or severe weather on markets.
4. **Agentic AI and LLM-powered investment assistants (11)** Build AI agents that research companies, summarize financial documents, interpret news, generate explanations, recommend trades, or support portfolio decisions. Several ideas involved connecting an agent to Binance, MetaTrader, APIs, or paper-trading systems.
5. **Multi-asset and alternative-market applications (10)** Apply financial analytics to crypto, forex, CFDs, options, commodities, energy markets, ETFs, or digital assets. Examples include crypto trading bots, forex hedging systems, options strategy backtesting, gold-price analysis, and energy-price forecasting.

Course Objectives vs. Expected Outcomes

(1 “easy” → 7 “hard”)

- 1 [Skills] Replicate notebooks and code: Python, analytics, and finance skills
- 2 [Skills] Register with a broker, transfer funds, and practice small trades
- 2 [Leaderboard] Compete to complete all homework assignments and open-ended questions
- 3 [Portfolio] Develop a project to generate trade recommendations
- 4 [Investing] Customize code for your market, product, or idea; experiment with models
- 5 [Potential Profits] Apply models to paper or real-money trading
- 6 [Long-term] Achieve sustained profitability; track trades, debug predictions, improve models
- 7

Data-Driven Decisions

Making informed choices for
personal savings and
investments

Let's start with macroeconomics and the first data source (FRED)

Real potential GDP is the CBO's estimate of the output the economy could produce if its capital and labor resources were used at a high rate. This data is adjusted to remove the effects of inflation. **The prediction is stable: ± 2 -2.3% YoY last 5 years**

Code snippet 1

- [About the FRED datasource](#)
- [About the metric US GDPPOT \(Real Potential Gross Domestic Product\)](#)
- [\[Simply Wall St.\] Macro update Q1'25 \(tariffs and global valuation are important\)](#)
- [2024 Article about the US Economy on PythonInvest](#)

- [Q1'25] Tariff and price bump, global valuation
- SNP500 Q1'25 earnings insights: https://advantage.factset.com/hubfs/Website/Resources%20Section/Research%20Desk/Earnings%20Insight/EarningsInsight_041725.pdf

Do I save faster than inflation? “Core CPI” (US) is 2.5%

(Jul'25) The **Consumer Price Index for All Urban Consumers: All Items Less Food & Energy** is an aggregate of prices paid by urban consumers for a typical basket of goods, excluding food and energy. This measurement, known as "**Core CPI**", is widely used by economists because food and energy have very volatile prices. **"Core CPI" declined from 3.7% on Feb'24 to 2.5% on Jul'26**

Code snippet 2

- **Option 1 [used in the code]:** Monthly "Core CPI" Index on FRED: <https://fred.stlouisfed.org/series/CPILFESL>
- Option 2: Daily inflation estimate from financial markets: <https://fred.stlouisfed.org/series/T10YIE>
- Option 3: Daily "ex-ante" interest rate (cost of borrowing money) <https://fredblog.stlouisfed.org/2022/05/constructing-ex-ante-real-interest-rates-on-fred>

[Ireland] Saving money with cash interest may bring you 2–3.1% interest in 2026, lower than inflation (3.4%)

- **Currency:** USD or EUR (harder when income vs. investments in different currencies).
- **Inflation** in Ireland is [3.4%](#) and in the US is 2.5% (Jul'26)
- **Strategy 1: Do nothing (0% interest, full liquidity)**
- **Strategy 2: Bank Savings Accounts:** For example, the traditional bank interest rate in Ireland is 2–3.1% in EUR ([comparison](#)). It has a money protection scheme.
- **Strategy 3: Money Service Businesses (MSBs)** offer in Aug 2026: **2.02%** ([Wise](#), Apr'25 - 2.24%, Apr'24 - 3.67%) in EUR and **3.44%** in USD ([Wise](#), Apr'25 - 4.08%, Apr'24 - 5.05%)
- **Strategy 4: Broker Accounts:**
 - (NL) DEGIRO pays **0%** interest on uninvested cash, regardless of currency
 - (DE) [Trade Republic](#) bank/broker offers **3% interest** on uninvested EUR cash (down from 4% in 2024). **Trade Republic** is a licensed German bank with deposit protection up to €100,000
 - (US) [Interactive Brokers](#) offers up to **3.1%** in USD (4.1% in 2025, 4.83% in 2024), and up to **1.7%** (2.7% in 2025, 3.45% in 2024) in EUR (for cash balances above \$10,000/€10,000; 0% for the first \$10,000/€10,000)
 - (IE + EU) **Revolut** [Flexible Cash Funds](#) (money Market): **2.1%** EUR, **3.69%** USD (2.59%, 4.36% in 2025)

If you start investing (in the US), the most important rate is Fed Funds Rate. Generally you can get slightly higher rates of return on debt instruments (4-5% in USD in Apr'25).

The **Federal Funds Rate** is the interest rate at which depository institutions trade federal funds (balances held at Federal Reserve Banks) with each other overnight. The latest value is **3.63%** in 2026 (4.33% in 2025, 5.33% in 2024) one year ago.

Code snippet 3

- **Fed Funds Rate** is **3.63%** (Jan-Aug'26), which is higher than cash rate (2-3%) : <https://fred.stlouisfed.org/series/FEDFUNDS> ;

- There is [ongoing debate](#) about future Fed rate changes, with markets expecting more rate cuts in 2026
- U.S. Treasury securities are considered the lowest risk ("**risk-free rate**" - **typically it is a 3 month t-bill (3.88%) for short-term and 10Y treasury yield for long-term (4.74% in Aug'2026)**). The Treasury [yield curve](#) (1 month to 30 years) is available daily. It can be reconstructed from [FRED time series](#), like [DGS1](#) (1MO) to [DGS30](#) (30Y). Longer-term bonds usually offer higher yields, but sometimes the [yield curve inverts](#) (long-term yields fall below short-term yields).
- **Bond yield** $\approx$ "risk-free rate" + premium for risk (e.g., default). Corporate debt and riskier bonds can yield **4–6%** or more in 2026, but with higher risk. Data: Moody's Corporate Bond Yield Averages series on FRED ([AAA](#) is 5.8%, [BAA](#) is 6.2%)

3M t-bills: <https://fred.stlouisfed.org/series/TB3MS>

Bonds 10Y quality: <https://fred.stlouisfed.org/series/HQMCB10YR>

Look at Index Growth for a Benchmark When Investing in Stocks

The **S&P 500 Index** tracks 500 leading U.S. companies, offering a snapshot of the country's stock market health. It is a go-to benchmark for investors and reflects a diverse range of sectors.

Code snippet 4

- Individual stocks can deliver a much wider range of returns-both positive and negative-due to higher risk - check the [heatmap](#)
- **The S&P 500** is the most widely recognized benchmark for U.S. stock performance.
 - From 1928 to 2026Q1, the average annual return (with dividends reinvested) was **10.09%**; inflation-adjusted, the real return was 6.81% per year [\[Source\]](#)
 - The **index's performance can vary widely** from year to year. For example, in 2024 the S&P 500 gained 23.31%, in 2025 it is down -6.06% (as of April 26, 2025), and is +20.47% in 2026 (as of August 24, 2026)

“Rule of 72” Concept: How Quickly Can You Double Your (Real) Capital?

- How Long Does It Take For Your Money to Double? <https://www.visualcapitalist.com/visualized-how-long-does-it->

<https://www.visualcapitalist.com/visualized-how-long-does-it-take-to-double-your-money/>

The long-term success of a profitable strategy is ... to be higher than a benchmark

- **Positive return:** Ideally, any strategy should have a rolling positive return over any selected period of 6–12 months.
- **60/40 portfolio** (equity/bonds): Historically, a global 60/40 portfolio has offered a lower yield, with recent baseline forecasts around **7.1%** nominal (about **5%** inflation-adjusted)
- **S&P 500 (Equity):** Over 30 years (1995–2025), the average annual return was **9.9%** (about 6.3% inflation-adjusted). There are years with declines of –20% to –40% in a single year ([link](#))
- Sweet spot: The “sweet spot” for long-term returns is typically in the range of **6–10%** per year, net of inflation.
- Warren Buffett (Berkshire Hathaway): Achieved a compound annual growth rate (CAGR) of **~20%** over several decades ([LINK](#))
- Renaissance Technologies (Medallion Fund): Achieved a reported CAGR of **66%** (gross, before fees) over three decades, using highly leveraged quantitative strategies : [LINK](#), [LINK2](#)

Another research from Goldman Sachs on 60/40 portfolio and 2024: <https://www.goldmansachs.com/intelligence/pages/the-60-40-portfolio-should-offer-a-better-risk-reward-in-2024.html>

<https://www.goldmansachs.com/insights/articles/> – Tariffs and US exch rate

CAGR definition: <https://corporatefinanceinstitute.com/resources/valuation/what-is-cagr/>

#:~:text=The%20CAGR%20formula%20is%20equal,of%20Periods)%20%E2%80%93%201.

Top 25 stocks in S&P500 Apr'24: <https://www.investopedia.com/best-25-sp500-stocks-8550793>

Concept: Generic Risk-Reward Curve by Asset Class

The Efficient Frontier

- Chart adapted from: Explaining the Investing Concept of Risk and Return ([link](#))

- [How asset classes performed in the first half of 2026?](#) (Fidelity research)
- PythonInvest's articles on Portfolio Optimization: "[All Weather Portfolio with Crypto](#)", "[Practical Portfolio Optimisation](#)"

<https://europe.ark-funds.com/insights/big-ideas/>

Select your risk to define one of the major “characteristics” of a future strategy

- **Risk tolerance** varies by investor: Each investor has *a unique risk-return profile*, shaped by their financial goals, investment horizon, and personal comfort with potential losses ♦
- **General guidance:** It is often recommended to take *more risk earlier* in your investment journey (when you have less capital and a longer time horizon), and *shift to more conservative strategies* (such as bonds or diversified ETFs) as your capital grows or your goals near ♦
- **Defining success:** Success is easier to measure relative to a benchmark-such as the risk-free rate or a passive investment portfolio. *Common metrics include the Sharpe ratio, Sortino ratio, and maximum drawdown* ♦
- **Data perspective:** Because all markets are interconnected, *you will likely need data on various asset classes and macroeconomic factors*-even if your strategy focuses on a single asset type (for example, US large-cap stocks)

Practical setup

Hosted Colab vs. Jupyter
Notebook (Local) vs. IDE

Google Colab is the main environment for the course

Why (Hosted) Colab? ♦ [Google Colab](#) lets you open existing code (File → Open → GitHub) or create notebooks from scratch.

- **[+] No barriers to start:** No installation required; runs in your browser.
- **[+] Build-in AI features (Gemini 2.5/3 Flash)**
- **[+] Pre-installed libraries:** Most popular Python, data science, and machine learning libraries (e.g., NumPy, Pandas, TensorFlow, PyTorch) are ready to use.
- **[+] Free GPU/TPU access:** Accelerated hardware (e.g., T4/L4 GPU) included for faster computation.
- **[+] Easy saving and sharing:** Notebooks are saved to your Google Drive and can be shared or collaborated on in real time.
- **[+] Dynamic visualizations:** Supports interactive graphs and outputs.
- **[+] Cloud-based:** Work from any device, anywhere; autosaves your work
- **[-] Sessions may disconnect,** especially during long or resource-intensive tasks.
- **[-] You may need to re-upload files** or reinstall dependencies after a session ends.
- **[-] Free tier has usage limits;** paid plans (Colab Pro/Pro+/Pay As You Go) offer more resources and longer runtimes

Alternative 1: (Local) Jupiter Notebooks and Anaconda: full local control and no disconnects ♦ **Alternative 2:** IDE (e.g. VS Code) : Can do the AI-assisted dev., can connect to Jupyter/Colab kernels for hybrid dev. ♦

Quick Links

- Short Colab intro on PythonInvest: <https://pythoninvest.com/long-read/python-environment>
- Paid plans for Colab (Pay As You Go vs. Colab Pro vs. other) : <https://colab.research.google.com/signup>

Data Sources For Stocks

An extensive list of data sources for equities and more

- **OHLCV Data** (Yahoo Finance, Massive.com)
- **Technical Indicators** (covered in Module 2; available via Yahoo Finance, TA-Lib, or built-in libraries)
- **Macroeconomic Data** (FRED, Pandas DataReader, Eurostat, etc.)
- **Financial Reporting** (SEC EDGAR: 10-K, 10-Q, 8-K filings)
- **News** (Massive.com, NewsAPI)
- **Fundamental Data** (Yahoo Finance, paid data providers, or web scraping)
- **Alternative Data** (e.g., web traffic, YouTube revenue, satellite imagery, social media sentiment from X/Twitter, Reddit, Glassdoor, app usage, credit card data)
- **Events** (Earnings calendars, ETF flows, activist investor actions, mergers & acquisitions, conference calls)

Feel the data: start from Stocks Screener (TradingView example)

- Many data providers offer free stock screeners; for example: <https://www.tradingview.com/screener/>
- Use “hand” rules to quickly filter stocks, such as: select all stocks in the **S&P 500** with a **dividend yield > 2%**, **revenue growth > 0% YoY**, and a **price-to-earnings growth (PEG) ratio < 0.5** and **Upcoming earnings date =Next Week**
- Here is the recent article on PythonInvest: <https://pythoninvest.com/long-read/stock-screening-using-paid-data>

[Main Data Source] OHLCV Data from Yahoo Finance

Code snippet 5

OHLCV data-comprising Open, High, Low, Close, and Volume-forms the **foundation of informed trading** and investment decisions. By analyzing these data points, traders and investors gain insights into price movements, volatility, market sentiment, and liquidity. These insights help assess risk, identify trends, and execute trades with greater confidence.

- Library folder on PyPi: <https://pypi.org/project/yfinance/>
- Recommended reading:
 - [yfinance Library - A Complete Guide](#) [Algotrading 101]
 - [Yale Lectures \[updated 2024\]](#)
- Data available via yfinance:
 - Periodical: minute, hourly, daily, weekly, monthly data
 - Different Asset types: [indexes](#), [bonds](#), [most-active](#) and other stocks, [crypto](#), [top ETFs](#), [currencies](#), [options chain](#) (not historical)
 - Dividends
 - [Fundamental data](#): (Latest info) about company, (4 years) of financial reporting

Paid data (Massive.com, etc.)

Paid data services like Massive.com, Yahoo Finance (official) offer distinct advantages over free sources. Premium providers deliver **real-time, high-quality** data with **greater accuracy, reliability**, and **broader market coverage**. They often include granular market data, advanced analytics, and tools

- **Free Tier/Trial**: Most paid data services offer a free tier or trial period: e.g. Massive.com can deliver 5 calls per minute for free: <https://massive.com/pricing> (\$29/month min paid tier)
 - Article about Financial News on PythonInvest using FREE TIER: <https://pythoninvest.com/long-read/chatgpt-api-for-financial-news-summarization>
- [Massive.com] Sometimes has unique data like [“Stock Fundamentals Ratios”](#) or News by ticker

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Macroeconomics stats

Code snippet 6

Macro data provides big-picture insights into **economic trends**, guides **policy decisions**, influences **market sentiment**, evaluates **sector performance**, and offers **global perspectives**. Traders and investors rely on it to make informed decisions, anticipate market movements, and manage risks effectively.

- [US] [FRED: Federal Reserve Economic Data](#)
 - Many useful charts with explanations about time series.
 - Can be Daily, Weekly, Monthly, Quarterly, sometimes you need to transform the data (e.g. calculate growth period-over-period)
 - For other countries: usually freely available from statistical/government agencies in other countries
 - Can be downloaded via [Pandas Data Reader \(PDR\)](#), or [Nasdaq's Quandl](#), or directly from the website (CSV) 💡
- There are many other data sites with (global) macro stats, for example:
 - [US indicators - latest dates] <https://tradingeconomics.com/united-states/indicators>
 - [Global interest rates] <https://tradingeconomics.com/country-list/interest-rate>
 - [DE stock market and related] <https://tradingeconomics.com/germany/stock-market> 💡
- Example macro stats
 - Financial markets benchmarks, consumer/investor behaviour, and economic cycle indicators
 - Other examples: unemployment, inflation, central bank interest rates (Fed, ECB, BoE, etc.), savings rate, debt rate
 - Asset benchmarks: gold/silver, commodities, crude oil, real estate indices, etc.
 - See the article on PythonInvest for 60+ macro indicators: <https://pythoninvest.com/long-read/macro-indicators-affecting-stock-market>

Financial Reporting for Public Companies

Code snippet 7

Financial reporting provides transparency, accountability, and credibility to stakeholders by offering accurate insights into a **company's financial performance and position**. It enables informed decision-making, enhances investor confidence, and facilitates access to capital markets. **Compliance with regulatory standards** ensures consistency and reliability in financial information, supporting effective risk management and stakeholder relations.

Data:

- [What is Edgar?](#) - millions of filings to SEC for free
- 10-K and 10-Q (SEC's filing or can be downloaded from company's IR page)
- [FREE access: raw txt files from Sec] <https://gist.github.com/anshoomehra/ead8925ea291e233a5aa2dcaa2dc61b2>
- Some free/paid APIs on top of the free tier : sec-api.io ([article](#), [pricing](#)) https://medium.com/@jan_5421/how-to-download-and-scrape-10-k-filings-from-sec-edgar-b0d245fc8d48
- <https://sec-api.io/resources/extract-textual-data-from-edgar-10-k-filings-using-python>
- <https://pypi.org/project/python-edgar/>

Web Scraping

Code snippet 8

Web scraping in finance is crucial for accessing **real-time data** from diverse online sources, **including news sites, social media, and forums**. This data provides valuable insights into market sentiment, trends, and events, aiding traders in making informed decisions and identifying potential opportunities. By leveraging web scraping techniques, traders can stay ahead of market developments, gain competitive intelligence, and enhance their trading strategies effectively.

- **Easy option:** use `pandas.read_html()` for static web pages. Example: <https://companiesmarketcap.com/most-profitable-companies/>
- **Advanced scraping:** For interactive sites (e.g., those requiring button clicks or logins), use tools like BeautifulSoup or Selenium. These allow you to automate browser actions and extract dynamic content. - example long-read on PythonInvest: "[Scraping EPS](#)" ([example page for scraping](#))
- [Trick] Sometimes even CSV is available:

- Articles on EPS (fundamental data)
 - Idea: the most important fundamental indicator. "Surprise" is important for a stock jump
 - <https://www.macrotrends.net/stocks/charts/F/ford-motor/eps-earnings-per-share-diluted>

Financial News - NewsAPI, Massive.com API

News fuels financial trading by providing critical insights into **market-moving events**, economic indicators, and **corporate developments**. **Timely access** to news helps traders anticipate market trends, assess risks, and capitalize on trading opportunities. By incorporating news analysis into their strategies, traders gain a competitive edge and enhance their ability to make informed decisions in dynamic markets.

- [NewsAPI.org](#)
 - Has free tier, but lagging 24 hour and only 1 month of history. PythonInvest article on [News Sentiment](#)
- Massive.com API
 - 5 calls/minute and 2 years history is free. Check [Weekly News Feed](#) on PythonInvest and "[Leveraging OpenAI's API for Financial News Summarization](#)" long-read
- [comment from Ivan] News are biased towards largest companies. It's hard to get real-time news information, can be messy and hard to extract trading signals, not very deep in the analysis. Although I like some sources like CNBC (partially free) or FT.com (paid)

Other Related Data Sources and Alternative data

Alternative data offers unique insights **beyond traditional sources**. It encompasses diverse datasets like **satellite imagery, social media sentiment, and consumer transactions**. This data **complements traditional analysis**, providing deeper understanding, **early signals**, and competitive edges. Traders and investors leverage alternative data to uncover hidden trends, validate hypotheses, and gain a strategic advantage in the market.

- **Event driven strategies on the related data:**

- 2026 Idea : Tokenized Stocks ([CoinMarketCap](#) link)
- IPOs
- Major conferences and announcements
- Large contracts
- M&A activity involving private companies
- Field research (e.g., health, biotech, machine learning & AI developments)

- **Other Sources**

- Trends & Social media: [Google Trends](#), StockTwits, Reddit, X (Twitter), etc.
- ETF Fund flow - e.g. <https://hedgefollow.com/funds/Renaissance+Technologies> or [Activist Investor cases](#)
- YouTube calls analysis - Analyze transcripts and Q&A from earnings calls to extract management sentiment or detect key themes
- Regulatory data (DOJ's antitrust filings) for IT? <https://www.justice.gov/atr/antitrust-case-filings>
- Satellite imagery: monitor physical assets like oil storage tanks, parking lots, consumer behaviour
- ...

Cheat Sheet for Data Selection in Your Project

- Select a market or country (or go global - for example, when you choose crypto).
- Choose benchmarks for comparison (indexes and/or other asset classes).
- Select macroeconomic indicators (global or regional).
- Dataset size: How many tickers/companies do you want to include? What time period and history do you need?
- Fundamental data: Do you need company fundamentals? (e.g., Yahoo Finance provides 4 years of data)
- Financial reporting and alternative data: Will you use financial reporting data (e.g., SEC filings) or alternative data sources?

IMPORTANT: DO NOT "LEAK" DATA. You must use only the data available up to the prediction date-never use information from the future in your model or analysis.

RECAP

- Financial rate of return
- Developing an investment idea
- Macroeconomic data and

What we have learned?

indicators

- Stock datasets and data sources
- Cheat sheet for starting a project

Generic Recommendations for the Course

- **Be active:** ask questions (but not too many) and contribute to discussions.
- **Prefer the simplest option when possible and build on it:** e.g. reuse class materials and try to update it with new features/models/strategies
- **Home Assignments** are designed to be straightforward, but open-ended questions are included to encourage you to find your own unique approach.
- **Start working on the project early** to allow time for multiple iterations and improvements.
- **Leverage your edge:**
 - Build new skills and a portfolio, especially if you are just starting out.
 - Consider focusing on the local market or a vertical you already know well (where you work or live).
 - Use advanced data extraction, automation, or technical stacks if you have a software engineering background.
 - Apply better models if you are an ML practitioner.
 - Explore advanced strategies if you are a trader.
 - Spend more time on the course if you are a student.

Module 1

- **HA1:** 5 questions + 2 extra
- **Deadline:** submit within 3 weeks (by September 14, 11:59pm GMT+1)
- **Designed** to be straightforward and relatively easy. You can provide feedback in Telegram/Slack (under the HA announcement) or when you submit the HA.
- **Get extra points** for posting your homework and tagging #PythonInvest and #Datatalks.club on any public platform (e.g.

Medium, LinkedIn, X (Twitter), etc.).

We've answered some questions about the course format, leaderboard, Slack community, etc ? → Check out the [\[2026\] Pre-Launch Q&A session](#), where we answered many questions about the course.